BSc (Hons) Food Science and Nutrition			
Code	**Module**	**Module Description**	**Credits**
AP0616	Biological and Food Sciences Research Project	This module guides students in planning a literature review on a specific research topic, using relevant sources and databases to gather information, and critically analyzing primary literature. It also covers evaluating experimental protocols, safely conducting experiments, and discussing data validity. Students will present their findings through an oral presentation and a written report.	40
AP0617	Product Development and Sensory Analysis	This module covers the theoretical foundations of food market research and the innovation and development of new food products, with a focus on current and future industry trends. Students will engage in a group research project to apply practical skills in market research and new product development. Theoretical knowledge of food ingredients and consumer drivers will be used in a practical assignment to create innovative products. Additionally, students will study sensory analysis, data analysis, and statistics, along with gaining practical experience in Good Manufacturing Practices, New Product Development workflows, and conducting product trials.	20
AP0618	Food and Nutrition - Policy and Issues	This module focuses on reflecting current issues in the dynamic field of food science and nutrition. The content adapts to contemporary developments, covering the scope of food policy, key influencers, and the factors shaping it. Students will explore policy in the food system from production to consumption, examining global issues like the Millennium Development Goals and food waste's environmental impact. The module also reviews current policies and their development processes, with a focus on food, public health, and political influence in the UK, EU, and globally. It further enhances understanding of trends in a consumer-driven food system, considering economic and social	20

		factors, public health, and the media's role in communicating food-related information.	
AP0622	Investigative Sport and Exercise Nutrition	This module aims to develop an understanding of the principles and applications of nutritional interventions in the context of sport and exercise performance and health outcomes. The students will learn about the scientific evidence behind health claims associated with dietary interventions, the use of ergogenic aids and other supplementation through critical evaluation. The module also covers the effects and detection of drugs and doping agents within sport. In addition, the role of the nutritionist in practice will be reviewed in accordance with external guidelines/codes of conduct for nutrition professionals.	20
AP0640	Clinical Nutrition	This module is designed to develop an understanding of the principles of clinical nutrition, dietary intervention and treatment strategies. The students will learn about the biochemical basis of a range of disorders that are encountered in nutrition or related practice. The module also covers assessing nutritional status using anthropometric, clinical and biochemical data and how these relate to case studies and nutrition-related to disease. It highlights how each clinical or metabolic disorder influences nutritional status, and the underlying metabolic complications that cause these conditions will be highlighted. Additionally, students will learn about nutrition principles, dietary management, and control in clinical settings, including how clinical conditions impact nutritional status and are managed through proper nutrition.	20

BSc (Hons) Biomedical Science			
Code	**Module**	**Module Description**	**Credits**
AP0606	Biomedical and Biological Sciences Research Project	This module enables the students to design a literature investigation on a specific research topic and utilize relevant sources and databases to gather information. The module also covers conducting experiments safely and effectively, and discussing the validity and importance of the results. Additionally, the students will present their research findings through an oral presentation and a written report in the appropriate style and format.	40
AP0607	Molecular Cell Interaction	This module teaches about mammalian cell signaling processes correlated with their role in disease pathogenesis and cellular responses to toxic compounds. The module provides an understanding of molecular mechanisms underpinning carcinogenesis, molecular basis of bacterial signaling, bacterial pathogenesis, therapeutic strategies designed to alleviate disease/pathogenesis and the molecular basis of therapeutic design including drug action, chemotherapy and gene therapy.	20
AP0608	Pathology in Practice	This module represents the final stages of bringing together the knowledge and experience accrued from previous modules of the Pathology Specialties (Microbiology/ Immunology/ Cellular pathology/ Haematology/ Clinical Biochemistry). You will learn how to examine and assess the clinical symptoms and resulting results and data presented as a case study. Module Synopsis As a group, your team will decide the appropriate testing regime, in order to reach a diagnosis. This will take the form of preliminary investigations that should point the team in the right direction. Comprising of group discussions and guidance from tutors in the assessment of demonstrations/practical work, and data acquisition, your team will collectively contribute data,	20

		knowledge, and understanding of the test results, to unravel the complexities of the case.	
AP0609	Advanced Analytical Techniques	This module covers advanced analytical techniques in modern Biomedical Sciences, focusing on their scientific principles and applications in medical diagnostics and research. Students will learn about genetic engineering, recombinant protein purification, flow cytometry, Chromatin immunoprecipitation (ChIP), automated enzyme analysis, next-generation sequencing, and microarray technologies. The module combines theoretical learning with hands-on laboratory experience, emphasizing the evaluation and selection of appropriate analytical methods and experimental design, crucial for future careers.	20
AP0610	Genomics	This module explains the application of computer-based tools in the management and analysis of biological data and it will enable you to gain an understanding and practical experience of state-of-the-art bioinformatics techniques. This focuses on the application of bioinformatics to eukaryotic genomes and human genome and its analysis using microarrays and next generation sequencing techniques. You will gain experience in CRISPR-Cas applications, Gene set identification and annotation with machine learning concepts related to computing.	20
AP0611	Drug Design and Development	This module explores the drug discovery and development process, beginning with the selection of a disease, identifying disease targets, and determining the appropriate molecules to use. The module starts with a discussion of drugs and drug testing in a historical and social context, followed by an understanding of how drugs interact with their targets in the body. You will also learn how to assess drug properties, identify lead compounds, and develop them into clinically approved drugs. Topics of the module include pharmacogenomics, emerging technologies in personalised medicine, principles of clinical trials and	20

		pharmacological methods as well as a consideration of ethical issues.	

BSc (Hons) Biotechnology			
Code	**Module**	**Module Description**	**Credits**
AP0606	Biomedical and Biological Sciences Research Project	This module enables the students to design a literature investigation on a specific research topic and utilize relevant sources and databases to gather information. The module also covers conducting experiments safely and effectively, and discussing the validity and importance of the results. Additionally, the students will present their research findings through an oral presentation and a written report in the appropriate style and format.	40
AP0607	Molecular Cell Interaction	This module teaches about mammalian cell signaling processes correlated with their role in disease pathogenesis and cellular responses to toxic compounds. The module provides an understanding of molecular mechanisms underpinning carcinogenesis, molecular basis of bacterial signaling, bacterial pathogenesis, therapeutic strategies designed to alleviate disease/pathogenesis and the molecular basis of therapeutic design including drug action, chemotherapy and gene therapy.	20
AP0609	Advanced Analytical Techniques	This module covers advanced analytical techniques in modern Biomedical Sciences, focusing on their scientific principles and applications in medical diagnostics and research. Students will learn about genetic engineering, recombinant protein purification, flow cytometry, Chromatin immunoprecipitation (ChIP), automated enzyme analysis, next-generation sequencing, and microarray technologies. The module combines theoretical learning with hands-on laboratory experience, emphasizing the evaluation and selection of appropriate analytical methods and experimental design, crucial for future careers.	20
AP0614	Applied Bioinformatics and Post Genomics	This module explains the application of computer-based tools in the management and analysis of biological data and it will enable you to gain an understanding and practical experience of state-of-	20

		the-art bioinformatics techniques. This focuses on the application of bioinformatics to eukaryotic genomes and human genome and its analysis using microarrays and next generation sequencing techniques. You will gain experience in CRISPR-Cas applications, Gene set identification and annotation with machine learning concepts related to computing.	
AP0612	Impact of Science on Society	This module examines a range of contemporary bioscience and technology topics, focusing on the underlying research and evidence, the ethical, legal, and social implications (ELSI), how science is communicated to the public, and its impact on modern society. Topics will include modern biomedical practices such as genetic screening, the allocation of scarce life-saving resources, the use of animals in research, and biodiversity conservation. A case study approach will be used for topic delivery, alongside developing transferable skills in reading, research, writing, analysis, and presentation, all aimed at enhancing science communication.	20